

Understood. For interview preparation, the answers should sound like what a candidate would actually say in an interview — **short, direct, 2–5 sentences**, with a small real-world example where useful. No architecture diagrams, step flows, or tutorial-style explanation.

I'll cover **all the Week 6 topics from your image** in that format.

Week 6 — DataOps, Data Quality & Governance Interview Q&A

1. DataOps

Q1. What is DataOps?

Answer:

DataOps is an approach used to improve the speed, quality, and reliability of data pipelines through collaboration, automation, testing, and monitoring. It is similar to DevOps but focuses mainly on data. For example, a company may automatically test and deploy dbt models whenever developers push changes to Git.

Q2. What are the key practices of DataOps?

Answer:

The main practices are collaboration, automation, version control, automated testing, CI/CD, monitoring, and continuous improvement. These practices help teams release data changes faster without compromising data quality.

Q3. What is the difference between DevOps and DataOps?

Answer:

DevOps mainly focuses on software application development and deployment, whereas DataOps focuses on data pipelines, transformations, and data quality. DataOps also has to deal with issues such as missing data, duplicate data, schema changes, and data freshness.

Q4. Why is collaboration important in DataOps?

Answer:

A data pipeline usually involves data engineers, analysts, business teams, and sometimes governance teams. Collaboration ensures that everyone understands source changes, business rules, and data-quality requirements before changes are deployed.

Q5. Give a real-world example of DataOps.

Answer:

Suppose a data engineer changes a customer transformation in dbt. Instead of deploying it directly, the change goes through Git, automated tests, code review, and then deployment. After deployment, the pipeline is monitored for failures and data-quality problems.

2. CI/CD for dbt and Airflow

Q6. What is CI/CD?

Answer:

CI/CD stands for Continuous Integration and Continuous Delivery or Deployment. CI automatically

validates code changes, while CD helps move successfully tested changes into higher environments such as staging or production.

Q7. Why do we need CI/CD for data pipelines?

Answer:

CI/CD reduces manual deployment errors and ensures that pipeline changes are tested before reaching production. For example, if a dbt model contains an incorrect SQL statement, the CI pipeline can detect it before it is merged.

Q8. How do you implement CI/CD for dbt?

Answer:

We store the dbt project in Git and run commands such as dbt compile, dbt build, and dbt test whenever a pull request is created. If the tests pass, the change can be reviewed and deployed.

Q9. What commands are commonly used in dbt CI pipelines?

Answer:

Common commands are dbt deps, dbt compile, dbt run, dbt test, and dbt build. dbt build is commonly useful because it can execute models and associated tests together.

Q10. How is CI/CD used with Airflow?

Answer:

Airflow DAG files are stored in source control and validated before deployment. We can check Python syntax, DAG import errors, and unit tests before publishing the DAG to the Airflow environment.

Q11. Why should an Airflow DAG be tested before deployment?

Answer:

A DAG may have syntax errors, incorrect dependencies, missing packages, or invalid configurations. Testing prevents a broken DAG from reaching the production Airflow environment.

3. Automated Testing in Data Workflows

Q12. What is automated testing in data engineering?

Answer:

Automated testing verifies data and pipeline rules automatically whenever a pipeline or transformation runs. For example, we can automatically check whether customer_id contains NULL or duplicate values.

Q13. Why is automated testing important?

Answer:

Modern pipelines process millions of records, so manually checking data is not practical. Automated tests help identify issues early before incorrect data reaches reports or dashboards.

Q14. What are some common data tests?

Answer:

Common tests include NULL checks, uniqueness checks, accepted-value checks, referential-integrity checks, range checks, and freshness checks.

Q15. Can a pipeline succeed but still produce bad data?**Answer:**

Yes. A pipeline can technically complete successfully but still produce incorrect data. For example, an ETL job may succeed even though 30% of customer IDs are NULL, which is why data-quality testing is also required.

4. Data Quality**Q16. What is data quality?****Answer:**

Data quality refers to how reliable and suitable data is for business use. Good-quality data should generally be accurate, complete, timely, consistent, valid, and unique.

Q17. What is data accuracy?**Answer:**

Accuracy means the stored data correctly represents the real-world value. For example, if a customer's actual age is 35 but the database contains 45, it is an accuracy issue.

Q18. What is completeness?**Answer:**

Completeness checks whether all required information is available. For example, if customer email is mandatory but 20% of customer records have NULL email addresses, there is a completeness problem.

Q19. What is timeliness?**Answer:**

Timeliness means data becomes available within the required business time. If a daily sales report should be available by 8 AM but the data arrives at 11 AM, it is a timeliness issue.

Q20. What is consistency?**Answer:**

Consistency means the same data should have compatible values across different systems. If a customer is marked as USA in CRM but Canada in the warehouse, there is a consistency problem.

Q21. What is validity?**Answer:**

Validity checks whether data follows defined rules, formats, or allowed values. For example, if valid order statuses are NEW, SHIPPED, and DELIVERED, a value such as ABC is invalid.

Q22. What is uniqueness?

Answer:

Uniqueness ensures that values that are supposed to be unique do not appear multiple times. For example, duplicate `order_id` values would indicate a uniqueness issue.

Q23. What is the difference between accuracy and validity?**Answer:**

Validity checks whether the value follows an expected rule, while accuracy checks whether the value is actually correct. An age of 45 instead of the customer's real age of 35 may be valid but inaccurate, while an age of 250 would normally be invalid.

5. dbt Tests**Q24. What is testing in dbt?****Answer:**

dbt testing allows us to validate the quality and integrity of transformed data. Common dbt data tests include `unique`, `not_null`, `accepted_values`, and `relationships`.

Q25. What does the `not_null` test do?**Answer:**

It checks whether a column contains NULL values. For example, we can apply it to `customer_id` if every customer record must have an ID.

Q26. What does the `unique` test do?**Answer:**

It verifies that every value in a column is unique. It is commonly used for columns such as `customer_id`, `order_id`, and other business keys.

Q27. What does `accepted_values` do?**Answer:**

It verifies that a column contains only approved values. For example, an order status column may allow only `NEW`, `SHIPPED`, `DELIVERED`, and `CANCELLED`.

Q28. What does a `relationships` test do?**Answer:**

It checks referential integrity between related data. For example, every `customer_id` in the orders table should exist in the customers table.

Q29. What happens when dbt test fails?**Answer:**

dbt reports which test failed and which records violated the rule. In a CI/CD pipeline, we can use that failure to prevent incorrect changes from being deployed.

6. Raw and Transformed Data Testing**Q30. Should we test raw data?**

Answer:

Yes. Raw data should be tested because identifying problems close to the source makes troubleshooting easier. We can check schema, NULL values, duplicates, freshness, and expected record volumes.

Q31. What tests would you perform on transformed data?**Answer:**

I would test business rules, calculations, uniqueness, relationships, NULL values, and expected ranges. For example, a sales model could have a rule that total sales amount should not be negative.

Q32. What is the difference between raw-data testing and transformed-data testing?**Answer:**

Raw-data testing mainly validates the quality of incoming source data. Transformed-data testing verifies whether the transformation logic and business rules have produced the expected result.

7. Data Lineage**Q33. What is data lineage?****Answer:**

Data lineage shows where data originated, how it was transformed, and where it is finally consumed. It helps with troubleshooting, impact analysis, auditing, and understanding data dependencies.

Q34. What is technical lineage?**Answer:**

Technical lineage shows dependencies between technical objects such as source files, tables, columns, transformation jobs, and reports. It is mainly useful to developers and data engineers.

Q35. What is business lineage?**Answer:**

Business lineage explains how business information and KPIs are derived. For example, it can explain that Revenue on a dashboard is calculated using quantity multiplied by unit price.

Q36. Why is data lineage important?**Answer:**

It helps us understand the impact of changes and trace data issues back to their source. If a column is being removed, lineage can show which downstream tables and dashboards depend on it.

Q37. What is the difference between technical and business lineage?**Answer:**

Technical lineage focuses on technical dependencies between systems, tables, and columns. Business lineage focuses on the meaning of the data and how business metrics are created.

8. Monitoring and Logging**Q38. What is monitoring?**

Answer:

Monitoring continuously checks the health and performance of systems and data pipelines. For example, we can monitor whether a Dataflow job is running, failed, or experiencing high processing latency.

Q39. What is logging?**Answer:**

Logging records detailed events and errors generated by applications or pipelines. Logs are mainly used to understand what happened when a problem occurs.

Q40. What is the difference between monitoring and logging?**Answer:**

Monitoring tells us that a problem exists, while logs provide detailed information about why the problem occurred. For example, monitoring may show that a Dataflow job failed, while the log may show that the failure was caused by insufficient permissions.

Q41. What would you monitor in Dataflow?**Answer:**

I would monitor job status, throughput, processing latency, backlog, worker utilization, CPU usage, and errors. For streaming pipelines, backlog and latency are especially important.

Q42. What would you monitor in Cloud Composer?**Answer:**

I would monitor DAG failures, task failures, retry counts, execution duration, scheduler health, worker health, and environment resource utilization.

9. Alerting Policies**Q43. What is an alerting policy?****Answer:**

An alerting policy defines a condition that triggers a notification when something abnormal happens. For example, we can notify the support team when a Dataflow job fails or when CPU usage remains above 90%.

Q44. Why do we need alerting?**Answer:**

Without alerts, engineers may have to continuously check dashboards manually. Alerting allows the team to react quickly when failures or abnormal conditions occur.

Q45. What is a notification channel?**Answer:**

A notification channel determines where an alert notification is sent. Depending on the monitoring platform and configuration, this could include email or integrated incident-management systems.

10. Log-Based Metrics

Q46. What is a log-based metric?

Answer:

A log-based metric creates a measurable metric from matching log entries. For example, we can count how many times an ORDER_PROCESSING_FAILED error occurs in application logs.

Q47. When would you use a log-based metric?

Answer:

I would use it when an application-specific condition appears in logs but does not already have a built-in monitoring metric. The log entries can then be counted and monitored.

Q48. Give an example of a log-based metric.

Answer:

Suppose an Airflow task writes CUSTOMER_LOAD_FAILED whenever customer processing fails. We can create a metric that counts those log entries and generate an alert when the count becomes greater than zero.

11. Data Governance

Q49. What is data governance?

Answer:

Data governance is the set of policies, roles, standards, and controls used to manage organizational data properly. It ensures that data is secure, reliable, understandable, and used according to business and regulatory requirements.

Q50. What are the main pillars of data governance?

Answer:

Important areas include data ownership, security, data quality, metadata and cataloging, privacy, compliance, lineage, and data stewardship.

Q51. Why is data governance important?

Answer:

Organizations have large amounts of sensitive and business-critical data. Governance ensures that people know who owns the data, who can access it, what it means, and how it should be protected.

Q52. What is data ownership?

Answer:

Data ownership identifies who is accountable for a dataset. The owner is usually responsible for decisions regarding access, usage, quality expectations, and retention.

Q53. What is a data catalog?

Answer:

A data catalog is a searchable inventory of an organization's data assets. It helps users understand what datasets are available, their definitions, owners, schemas, classifications, and other metadata.

12. RBAC

Q54. What is RBAC?

Answer:

RBAC stands for Role-Based Access Control. Instead of assigning permissions individually to every user, permissions are assigned to roles and users are assigned to those roles.

Q55. Why is RBAC important?

Answer:

RBAC simplifies access management and improves security. For example, all analysts can receive an ANALYST_ROLE with read-only permissions instead of manually granting permissions to every analyst.

Q56. What is the principle of least privilege?

Answer:

Least privilege means giving users only the permissions they require to perform their job. For example, an analyst who only reads reports should not normally receive permission to delete production tables.

Q57. What is the difference between RBAC and direct grants?

Answer:

With direct grants, permissions are assigned directly to individual users, which becomes difficult to manage at scale. RBAC groups permissions into roles, making access easier to control and audit.

13. PII and Data Masking

Q58. What is PII?

Answer:

PII stands for Personally Identifiable Information. It includes information that can identify or relate to an individual, such as name, email address, phone number, address, and government identification information.

Q59. What is data masking?

Answer:

Data masking hides sensitive information from users who do not need to see the original value. For example, an analyst may see ****@company.com instead of the employee's complete email address.

Q60. What is dynamic data masking?

Answer:

Dynamic data masking displays different values depending on the user's permissions or role without necessarily changing the original stored value. HR may see the full email, while an analyst sees a masked email.

Q61. Why do organizations mask PII?

Answer:

Masking reduces unnecessary exposure of sensitive information. It supports privacy, least-privilege access, and compliance requirements.

14. Compliance

Q62. What is GDPR?

Answer:

GDPR is a European data-protection regulation governing how organizations collect, process, store, and protect personal information. It also provides individuals with rights regarding their personal data.

Q63. What is HIPAA?

Answer:

HIPAA is a US law that includes privacy and security requirements for protected health information. Organizations handling applicable healthcare information must implement suitable safeguards.

Q64. What is SOC 2?

Answer:

SOC 2 is an assurance framework used to evaluate an organization's controls related to areas such as security, availability, confidentiality, processing integrity, and privacy.

Q65. What is the basic difference between GDPR, HIPAA, and SOC 2?

Answer:

GDPR focuses mainly on personal-data privacy in Europe. HIPAA focuses on protected healthcare information in the United States, while SOC 2 focuses on evaluating an organization's internal controls for trust and security.

15. Data Stewardship

Q66. What is data stewardship?

Answer:

Data stewardship is the day-to-day management of data definitions, quality rules, metadata, classifications, and governance standards. A data steward helps ensure that data is properly understood and maintained.

Q67. What does a data steward do?

Answer:

A data steward may maintain business definitions, define data-quality rules, classify sensitive data, manage metadata, and coordinate the resolution of data-quality issues.

Q68. What is the difference between a data owner and data steward?

Answer:

A data owner is accountable for the data and normally has decision-making authority. A data steward handles many of the day-to-day governance and quality activities for that data.

16. Dataplex

Q69. What is Dataplex?

Answer:

Dataplex is a Google Cloud service used for data management and governance across distributed data assets. It provides capabilities around metadata, discovery, governance, data quality, and related data-management activities.

Q70. Why would you use Dataplex?

Answer:

Dataplex helps organizations manage and discover large numbers of data assets across Google Cloud. It is useful when teams need centralized metadata, governance, data-quality capabilities, and easier data discovery.

Q71. What kind of data sources can Dataplex work with?

Answer:

Dataplex commonly works with Google Cloud data assets such as BigQuery and Cloud Storage. It helps organizations manage metadata and governance across those distributed data environments.

17. Automatic Discovery

Q72. What is automatic discovery in Dataplex?

Answer:

Automatic discovery examines data assets and automatically collects metadata about them. This reduces the amount of manual work required to document datasets.

Q73. What information can automatic discovery identify?

Answer:

It can help identify metadata such as schemas, columns, data types, and information about available data assets, depending on the configured source and discovery capabilities.

Q74. Why is automatic discovery important?

Answer:

Large organizations may have thousands of tables and files, making manual documentation difficult. Automatic discovery helps keep metadata more current and makes data easier to find and understand.

Scenario-Based Interview Questions

Q75. Your sales pipeline completed successfully, but today's sales numbers are 60% lower than normal. What would you do?

Answer:

I would first check whether the source data arrived completely and on time. Then I would verify row

counts, NULL values, transformation logic, and recent source/schema changes. A successful pipeline only confirms execution; it does not guarantee that the data is correct.

Q76. A source suddenly starts sending duplicate order IDs. How would you handle it?

Answer:

I would add a uniqueness test on order_id and prevent bad data from being promoted downstream. I would also investigate whether the duplicates originated from the source or were introduced during ingestion.

Q77. Twenty percent of customer emails suddenly become NULL. What type of problem is this?

Answer:

It is mainly a completeness issue because required information is missing. I would compare it with historical levels and raise an alert if it crosses the accepted threshold.

Q78. Customer age is 250. What kind of data-quality issue is this?

Answer:

It is primarily a validity issue because the value does not satisfy a reasonable business rule. We can add a validation rule that age must fall within an acceptable range.

Q79. The customer age stored is 45, but the customer's actual age is 35. What problem is this?

Answer:

This is an accuracy issue. The value looks valid but does not correctly represent reality.

Q80. A report contains correct information, but yesterday's data is still being shown today. What type of issue is this?

Answer:

This is a timeliness or freshness issue. The data itself may be correct, but it was not updated within the required business SLA.

Q81. CRM shows a customer in the USA, while the warehouse shows the same customer in Canada. What issue is this?

Answer:

This is a consistency issue. I would identify which system is the authoritative source and then investigate where the value changed during ingestion or transformation.

Q82. An analyst can see employee salaries, but they do not need this information. What would you do?

Answer:

I would remove the unnecessary access and apply least privilege using RBAC. Salary columns could also be protected using column-level security or masking where appropriate.

Q83. HR needs to see employee email addresses, but analysts should not. How would you handle it?

Answer:

I would provide access based on roles and apply masking for users who do not require the original values. HR could see the full email while analysts see a masked version.

Q84. A developer changes a dbt model and it breaks production. How can this be prevented?

Answer:

I would introduce CI/CD with automated dbt compilation and testing before code is merged. Changes should go through a pull request, automated validation, and code review before deployment.

Q85. A dbt test fails in your CI pipeline. Would you deploy the change?

Answer:

Normally I would not deploy until I understand and resolve the failed test. A failed data-quality test may indicate that the code or underlying data does not meet the expected requirement.

Q86. An Airflow DAG works locally but shows a Broken DAG error after deployment. What would you check?

Answer:

I would check the Airflow logs for import errors, missing Python packages, incorrect imports, environment differences, and invalid DAG configuration. I would also ensure DAG validation is part of CI before deployment.

Q87. A Dataflow job fails every night, but the team notices only the next morning. How would you improve this?

Answer:

I would configure Cloud Monitoring and an alerting policy for the job failure condition. The support or engineering team should receive a notification immediately instead of manually checking the job the next morning.

Q88. Your application logs `PAYMENT_LOAD_FAILED` whenever processing fails. How can you monitor this?

Answer:

I can create a log filter for that error and, if metric-based monitoring is required, create a log-based

metric counting those occurrences. An alert can then be configured when the count exceeds the expected threshold.

Q89. A Dataflow streaming job is running, but its backlog continuously increases. What could be happening?

Answer:

The pipeline may not be processing data as fast as it is receiving it. I would check throughput, worker utilization, autoscaling, hot keys, processing latency, and downstream bottlenecks.

Q90. A business manager asks where the Revenue number in a dashboard comes from. How would you answer?

Answer:

I would use data lineage to trace the metric back through the warehouse and source systems. Business lineage would explain the revenue definition, while technical lineage would identify the tables, columns, and transformations involved.

Q91. A developer wants to delete a column from a production table. What should be checked first?

Answer:

I would check data lineage to identify downstream models, reports, pipelines, or dashboards using that column. This prevents an upstream schema change from unexpectedly breaking dependent systems.

Q92. Your organization has thousands of BigQuery tables and users don't know which tables to use. What would you recommend?

Answer:

I would introduce proper data cataloging and metadata management, potentially using Dataplex capabilities in Google Cloud. Users should be able to identify the dataset description, owner, schema, classifications, and other relevant metadata.

Q93. Thousands of files are being added to Cloud Storage, and manually documenting them is difficult. What can help?

Answer:

Automatic discovery can reduce manual metadata management by discovering supported data assets and extracting metadata such as schema information. This makes large data environments easier to search and govern.

Q94. A business team reports bad data, but no team accepts responsibility. How would governance help?

Answer:

Governance should clearly identify the data owner and data steward. The owner is accountable for the dataset, while the steward helps manage definitions, quality rules, metadata, and issue resolution.

Q95. An auditor asks who accessed employee salary information last month. What should be available?

Answer:

The organization should have appropriate audit and access logs. Those records should help identify which identity accessed the data, when it occurred, and what action was performed.

Q96. An employee leaves the company. How does RBAC make access removal easier?

Answer:

If permissions are managed through roles, we can remove the user's role or group memberships rather than searching for individual permissions across many data objects. This makes access revocation more consistent and manageable.

Q97. A new analyst joins the team. Would you grant permissions table by table?

Answer:

Normally no. I would assign the analyst to the appropriate analyst role, which already contains the required permissions. This is easier to manage and audit.

Q98. An order exists with a customer ID that is not present in the customer table. Which dbt test would you use?

Answer:

I would use a relationships test. It can verify that every customer ID in the orders data has a corresponding customer record.

Q99. Order status contains IN_TRANSIT, but only NEW, SHIPPED, DELIVERED, and CANCELLED are permitted. Which dbt test can detect this?

Answer:

I would use an accepted_values test on the status column. The test will fail when unexpected values appear.

Q100. Your pipeline is healthy, but users say they do not trust the data. What would you improve?

Answer:

I would focus on data-quality tests, freshness monitoring, lineage, ownership, and clear business

definitions in addition to technical monitoring. Pipeline health tells us whether processing is working, but trust also depends on whether the data is correct, timely, and understandable.